

TASK 3: PHARMACOVIGILANCE & ADR ANALYSIS

Drug Selected: Ibuprofen

Introduction

Pharmacovigilance is basically the process of checking medicines after they are being used by people. It helps us find side effects and other problems related to medicines. This is important because not every side effect can be found during the first stages of testing.

For this task, I selected ibuprofen. Ibuprofen is a common medicine used for pain, fever and inflammation. It is easily available and used by many people. Even though it is useful, it can cause some side effects, especially if it is taken in high amounts or for a long period of time.

The World Health Organization collects reports about suspected medicine side effects in its global pharmacovigilance system. These reports help experts notice possible safety problems. A report does not always prove that the medicine caused the problem, so doctors and researchers have to look at the full situation.

1. About Ibuprofen

Ibuprofen belongs to a group of medicines called NSAIDs, which means non-steroidal anti-inflammatory drugs. It is mainly used to reduce pain, fever and swelling. Its DrugBank identification number is DB01050.

It is commonly used for headaches, muscle pain, tooth pain, period pain, fever and other conditions where pain or inflammation needs to be reduced.

2. How Does Ibuprofen Work?

Ibuprofen works by blocking two enzymes called COX-1 and COX-2. These enzymes help the body produce substances called prostaglandins. Prostaglandins are involved in pain, fever and inflammation.

When ibuprofen reduces prostaglandins, pain and inflammation also become lower. This is why the medicine works well as a painkiller and fever reducer.

The problem is that some prostaglandins also have useful jobs. They help protect the stomach and support normal kidney function. Because ibuprofen reduces them too, it can sometimes cause problems with the stomach or kidneys.

3. Side Effects of Ibuprofen

Ibuprofen does not cause serious problems in everyone. However, some side effects are important to know about.

- **Stomach problems:** indigestion, stomach pain, ulcers and stomach bleeding can occur.
- **Kidney problems:** in some people, ibuprofen can affect kidney function or cause the body to keep extra fluid.
- **Heart and blood pressure:** long-term or high-dose use may increase blood pressure and can increase the risk of serious cardiovascular problems in some people.
- **Allergic reactions:** some people may develop a rash, swelling, itching or breathing difficulty.
- **Liver problems:** serious liver reactions are uncommon, but they can happen and need medical attention.

4. Who May Be at Higher Risk?

The risk of side effects is not the same for everyone. People who already have certain health problems may need extra care when using ibuprofen.

- People with a history of stomach ulcers or stomach bleeding.
- People with kidney disease, heart disease or high blood pressure.
- Older people or people with poor general health.
- People who take blood-thinning medicines, aspirin, steroids or other NSAIDs.
- People who are dehydrated, especially during illness.

5. How Can the Risks Be Reduced?

Ibuprofen can usually be used more safely when a person follows the correct dose and does not use it unnecessarily for a long time. Some simple steps can reduce the risk of problems.

- Use the lowest dose that is needed to control the pain or fever.
- Do not continue using it for a long time without medical advice.
- Do not take two different NSAID painkillers together unless a healthcare professional tells you to.
- Tell your doctor or pharmacist about other medicines you are taking.
- People with kidney, stomach or heart problems should ask for medical advice before using it.
- Get medical help if there is vomiting of blood, black stool, chest pain, severe swelling or difficulty breathing.

6. Reporting Side Effects

If a person thinks that ibuprofen has caused a serious or unusual side effect, it is important to tell a healthcare professional. Suspected side effects can also be reported through the proper national pharmacovigilance system. These reports can help health authorities identify safety signals and protect other patients.

7. Conclusion

Ibuprofen is a useful and common medicine, but like all medicines it has risks. It works by reducing substances involved in pain and inflammation. The same action can also affect the stomach, kidneys and cardiovascular system in some people.

The main lesson from this report is that a medicine should be used carefully. Taking the correct dose, avoiding unnecessary long-term use, checking other medicines and knowing the warning signs can help reduce the risks. Pharmacovigilance is important because it helps us keep learning about medicine safety after medicines reach the public.

References

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